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# A.C. Circuits: Series RLC, Impedance and Power Factor

*In a.c. work the resistor, inductor and capacitor do not simply add: their voltages peak at different instants, so impedances combine as phasors — and the power factor measures how much of the current actually does work.*

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## 1. Aim and theory

**Aim:** to study a series RLC circuit on single-phase a.c., to measure its impedance and power factor, and to observe the behaviour of single-phase and three-phase circuits and loads.

A pure resistor drops a voltage *in phase* with the current. An inductor's voltage *leads* the current by 90° and a capacitor's voltage *lags* it by 90°, so in a series circuit the inductive and capacitive drops oppose each other directly. With reactances $X_L = 2\pi fL$ and $X_C = 1/(2\pi fC)$, the impedance of the series combination is

$$Z = \sqrt{\left( R^2 + (X_L - X_C)^2 \right)} \quad \cdot \quad I = V / Z$$

$$\tan \varphi = (X_L - X_C) / R \quad \cdot \quad \text{power factor} = \cos \varphi = R / Z$$

where φ is the angle by which the current lags the supply voltage (φ positive when $X_L > X_C$, a *lagging* power factor; negative, i.e. current leading, when the capacitor dominates). Only the resistor consumes average power, so the wattmeter reads

$$P = V \cdot I \cdot \cos \varphi = I^2 R$$

At **series resonance**, $X_L = X_C$: the impedance falls to its minimum Z = R, the current peaks, and the power factor becomes unity. The resonant frequency is $f_0 = 1/(2\pi\sqrt{(LC)})$. In a balanced three-phase star load, line and phase quantities are related by $V_L = \sqrt{3} \cdot V_{ph}$ and $I_L = I_{ph}$, and the total power is $P = \sqrt{3} \cdot V_L \cdot I_L \cdot \cos \varphi$ — measurable with two wattmeters (the two-wattmeter method), whose readings add to the total power.

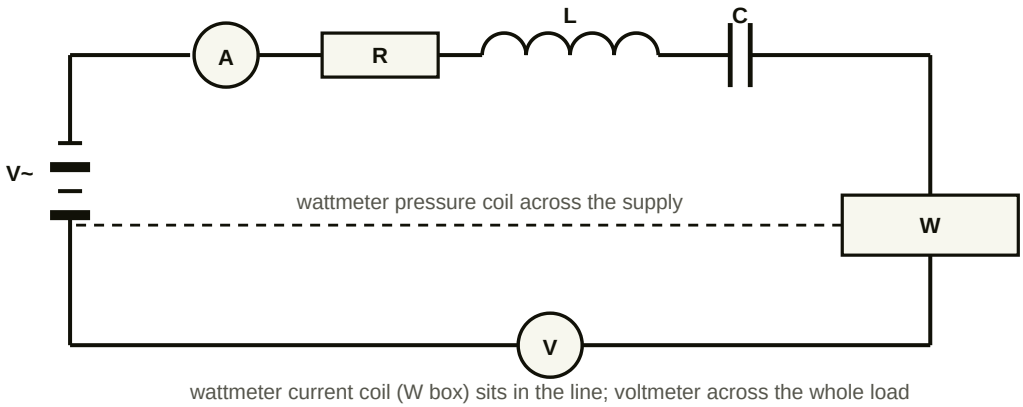



Figure 1 — Series RLC circuit with measuring instruments. The ammeter reads the common current; the wattmeter's current coil is in series and its pressure coil (dashed) across the supply, so it reads V·I·cos φ directly.

## 2. Apparatus and procedure

- **Apparatus:** single-phase a.c. supply (230 V, 50 Hz) with variac, resistor bank (or rheostat), iron-cored inductor, capacitor bank with selectable values, a.c. ammeter and voltmeter, wattmeter, three-phase supply and a balanced three-phase load for the second part.
- Wire Figure 1 with R = 30 Ω, L ≈ 0.1 H. Starting with the smallest capacitance, raise the supply to 230 V and record V, I and W.
- Step the capacitance upward through resonance (about 100 µF for this L) and beyond, keeping V constant, and record I and W at each step. Compute Z = V/I and cos φ = W/(V·I) for every row and compare with the values predicted from R, $X_L$ and $X_C$.
- Measure the voltages across R, L and C separately near resonance and confirm that $V_L$ and $V_C$ can each exceed the supply voltage while their difference stays small — the hallmark of series resonance.
- **Three-phase part:** connect a balanced star load to the three-phase supply; measure line voltage, phase voltage, line current and total power (two wattmeters). Verify $V_L/V_{ph} \approx \sqrt{3}$ and $P = W_1 + W_2 = \sqrt{3}\cdot V_L \cdot I_L \cdot \cos\varphi$.

| C (MF) | $X_C$ (Ω) | Z (Ω) | I (A) AT 230 V | REMARK |
|---|---|---|---|---|
| 25 | 127.3 | 100.5 | 2.29 | strongly capacitive, current leading |
| 50 | 63.7 | 44.0 | 5.22 | worked example below |
| 101 | 31.4 | 30.0 | 7.67 | resonance: Z = R, pf = 1 |
| 150 | 21.2 | 31.7 | 7.26 | net inductive, current lagging |

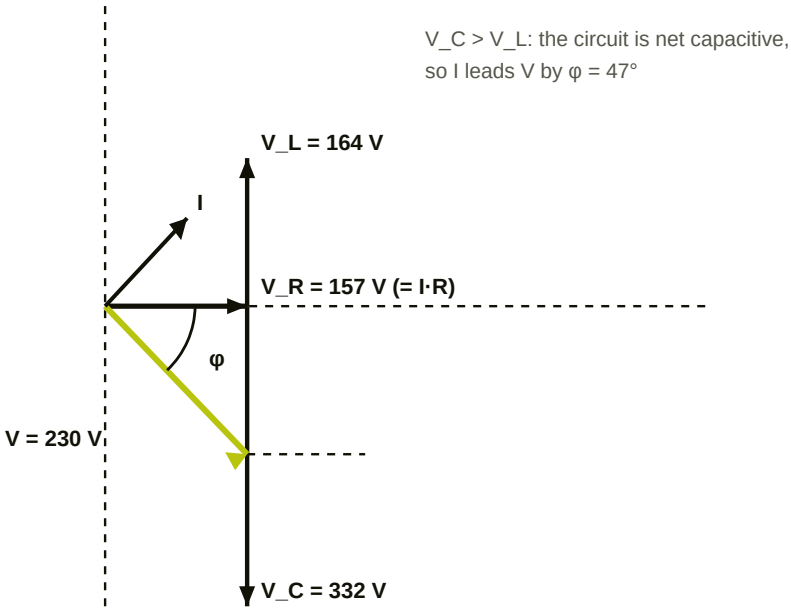



Figure 2 — Phasor diagram for the worked example. $V_R$ is drawn along the current; $V_L$ points 90° ahead of it and $V_C$ 90° behind. The supply phasor V is the vector sum — much shorter than $V_C$ alone, because the two reactive drops largely cancel.

## 3. Worked calculation

Series circuit on 230 V, 50 Hz: R = 30 Ω, L = 0.1 H, C = 50 µF.

$$X_L = 2\pi \times 50 \times 0.1 = 31.4\ \Omega \quad \cdot \quad X_C = 1 / (2\pi \times 50 \times 50 \times 10^{-6}) = 63.7\ \Omega$$

$$Z = \sqrt{(\ 30^2 + (31.4 - 63.7)^2\ )} = \sqrt{(\ 900 + 1040\ )} = 44.0\ \Omega$$

$$I = 230 / 44.0 = 5.22\ \text{A} \quad \cdot \quad \cos\varphi = R / Z = 30 / 44.0 = 0.68\ \text{leading}$$

$$P = V \cdot I \cdot \cos\varphi = 230 \times 5.22 \times 0.68 \approx 817\ \text{W} \quad (= I^2R = 5.22^2 \times 30 = 817\ \text{W}\ \checkmark)$$

The element voltages are $V_R = 5.22 \times 30 = 157$ V, $V_L = 5.22 \times 31.4 = 164$ V and $V_C = 5.22 \times 63.7 = 332$ V. They do *not* add arithmetically to 230 V; as phasors, $\sqrt{(157^2 + (332 - 164)^2)} = 230$ V exactly. The angle is $\varphi = \tan^{-1}((31.4 - 63.7)/30) = -47°$, the negative sign marking that the current *leads* — this circuit, with its oversized capacitor, behaves capacitively. A textile mill full of induction motors has the opposite problem (lagging current), and shunt capacitors are fitted precisely to pull such a load back toward the condition of this example and beyond, toward unity.

## 4. Precautions

- Near series resonance the voltages across L and C can rise to several times the supply voltage — check the voltage rating of the capacitor bank and the voltmeter range before approaching resonance.

- Iron-cored inductors are only roughly linear: their inductance falls as the core saturates, so keep the test current within the coil's rating and expect small disagreements with the ideal formulas.
- Discharge the capacitor bank through a resistor after switching off before touching the circuit — a charged 100 µF bank holds a painful, and on larger banks dangerous, amount of energy.
- For the wattmeter, confirm the pressure-coil connection (load side vs supply side) and correct for its small consumption if the load current is low.
- On the three-phase part, never open a current-carrying neutral or a CT secondary; check the phase sequence before energising rotating loads.

## Practice (viva preparation)

1. Define impedance and power factor. Why can impedances not be added arithmetically in a series a.c. circuit?
2. A coil of R = 20 Ω and $X_L$ = 15 Ω is connected to 230 V. Find Z, I and the power factor.
3. State the series-resonance condition and what happens to Z, I and cos φ at resonance.
4. In the worked example, $V_C$ = 332 V although the supply is only 230 V. Why is this not a violation of KVL?
5. How is the power factor of a balanced three-phase load found from two wattmeter readings?

### Answer key

1. Impedance is the total opposition to a.c. current, $Z = \sqrt{(R^2 + X^2)}$; power factor is cos φ, the ratio R/Z, marking what fraction of the apparent power is real power. Resistive and reactive drops are 90° apart in phase, so they combine by Pythagoras (as phasors), not by simple addition.
2. $Z = \sqrt{(20^2 + 15^2)} = 25$ Ω. I = 230/25 = 9.2 A. cos φ = 20/25 = 0.8 lagging.
3. $X_L = X_C$, i.e. $f = 1/(2\pi\sqrt{(LC)})$. Z is minimum and equal to R, the current is maximum for the given voltage, and cos φ = 1.
4. KVL holds for phasors, not for magnitudes. $V_L$ and $V_C$ are in antiphase and largely cancel; $\sqrt{(157^2 + (332 - 164)^2)} = 230$ V. Instant by instant, the drops still sum to the supply voltage.
5. Total power $P = W_1 + W_2$; with $V_L$ and $I_L$ measured, $\cos\varphi = P / (\sqrt{3}\cdot V_L \cdot I_L)$. (From the two readings alone, $\tan\varphi = \sqrt{3}\cdot(W_1 - W_2)/(W_1 + W_2)$.)